

GAURAV PAGARE

Machine Learning / AI Systems Engineer

gspagare03@gmail.com · +91 7264997994 · Pune, Maharashtra · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

SUMMARY

Founding engineer of an AI dubbing company's technical team. Built its speech-recognition training platform from an empty repository to distributed multi-GPU training running unattended on cloud, and its dubbing product from first commit to daily internal use. Comfortable end to end — data pipelines, DeepSpeed/NCCL debugging, Prefect/MLflow orchestration, provider integration, and the product surface on top. Now reports directly to the CEO.

EXPERIENCE

Ant Systemz Pvt. Ltd. — Software Engineer, ML Systems & Infrastructure

Apr 2025 – Present

Mumbai · First engineering hire; joined the day the technical team was formed and grew with it to ~10 engineers. · 428 commits across 9 repositories.

Speech recognition training platform (Whisper fine-tuning)

- Built an end-to-end ASR training platform from scratch — audio preprocessing (VAD via pyannote/Silero/Kaldi, Demucs separation, segment-level deduplication, augmentation, QC gates) through distributed training, checkpoint export and cloud sync.
- Productionized multi-GPU training on **DeepSpeed ZeRO-2** across 4-GPU nodes; diagnosed and fixed NCCL collective failures, torchrun TCPStore rendezvous hangs, per-rank CUDA device binding, distributed LR-schedule desync, and multi-GPU finalize deadlocks on GCP Blackwell instances.
- Orchestrated preprocessing and training with **Prefect + MLflow**; added GCS autoresume, hardware-aware GPU/CPU launch, a Docker Compose training operator, and pre-training readiness checks — enabling unattended multi-day runs.
- **Split a monolithic repository into two services**: extracted the preprocessing stack out of the training repo, restructured it into a stage-modular architecture with its own Prefect flows and MLflow dashboard, and reduced the training repo to training concerns only.
- Containerized the stack for **GCP / Vertex AI and AWS**, with regional Cloud Build push, CI smoke tests, and dual-cloud config parity.
- Versioned the data pipeline on GCS with incremental splits, race-condition-safe shared locks, and processed-version detection.
- Fine-tuned Whisper with **PEFT/LoRA**; fixed Hugging Face checkpoint export against live PEFT adapters and hardened decoder-label and validation contracts.
- Currently running preprocessing and training over a **200+ hour Marathi corpus**.

AutoDub → VOIT — AI dubbing platform

- Built the AutoDub speech pipeline — STT, regional STT/TTS, voice cloning and accent conversion — then merged it into the main dubbing application with user workspaces, projects, and a MySQL→PostgreSQL consolidation plan.
- Integrated and benchmarked speech/translation providers: **ElevenLabs, Qwen/DashScope, Deepgram, Cohere, OpenVoice, DeepDub**, plus **Shrutam-2 (STT)** and **Sooktam-2 (TTS)** for Indic languages; built dual-provider fallback so no single vendor is a hard dependency.
- Shipped a dual-provider **Voice Library** with emotion variants, per-line voice variants, accent-first builds, and self-healing repair.
- **Cut editor page load from 4,523 MB to 38 MB transferred (118x)** and from 13 GB to 176 MB of decoded resources, measured on a 7-minute 11-speaker multitrack job — by caching in-flight promises rather than resolved buffers, serving Opus previews instead of WAV stems, lazy-rendering waveforms via IntersectionObserver, and clamping a clip canvas being sized to 11,373,603 px (a ~3 GB allocation per clip, past the browser's limit).
- Traced a long-standing “re-dub plays the previous take” bug to a **function that did not exist** — two call sites guarded by typeof checks, so the only transcript refetch on those paths was a silent no-op for months.
- Eliminated regional-TTS hallucination through chunked synthesis; added speech-preserving timed placement, TTS duration capping, and same-language dub skip.
- Built an automated **dub QC** surface grading accent, dialect and TTS quality with an OpenAI audio model — replacing manual listening passes.
- Delivered the first successful **Iberian Spanish, Delhi-accent Hindi** and **UAE Arabic** voice clones for client samples.

Cloud, data and storage infrastructure

- Built multi-worker Google Drive → **S3 / Wasabi** migration and metadata-extraction services with threaded parallel transfer, S3 access points, and purpose-built IAM users, roles, buckets and policies; ran and monitored the production migration.
- Evaluated Wasabi against AWS for the real workload and recommended AWS-only — removing a redundant storage tier and its recurring cost.
- Produced corrected infrastructure cost sheets and a layered technical architecture document for external consultant review.

Earlier product work

- **Dubbing MVP (founding commit, Apr 2025)**: wrote the first commit and built the original product end to end — transcription, translation and TTS, Devanagari script handling, S3-backed staged storage with thread parallelism, AWS Lambda for audio extraction and auth, and migration to PostgreSQL.
- **Dub-Storage (Aug–Dec 2025, 97 commits)**: role-based enterprise media platform — folder hierarchies, cut-paste, restore, upload/download tracking, activity reporting, rate limiting; resolved database connection-pool exhaustion under load and enabled large-file uploads.

Technical leadership and ownership

- Onboarded and trained every engineer hired after me — all first-job hires — as the team grew to ~10.
- Served as the team’s primary technical point of contact: task breakdown, sprint planning, delivery follow-through and escalation. Reported directly to the CTO; **now reports directly to the CEO** with a weekly engineering report on delivery, blockers and decisions.
- Owned architecture and build-vs-buy decisions across storage, model providers, GPU/instance sizing and orchestration tooling.
- Led vendor relationships and commercial negotiations with ElevenLabs, Cohere, AWS, GCP and Akamai — covering pricing, credits and integration escalations.
- Trusted with independent purchasing authority for engineering tooling and AI/cloud subscriptions.

Whole New Level — Frontend Web Developer

Mumbai

Aug 2023 – Mar 2025

- Built and maintained 20+ WordPress sites, including the Communities Foundation of Texas (CFT) and CCRSM projects; implemented WooCommerce flows, site migrations and performance optimization.

Ahen Pvt. Ltd. — Frontend Developer Intern

Pune

Dec 2022 – Apr 2023

- Built responsive React interfaces to pixel-perfect designs within a function-based architecture.

SKILLS

ML & Speech PyTorch · DeepSpeed (ZeRO) · Hugging Face Transformers · PEFT/LoRA · Whisper · VAD (pyannote, Silero, Kaldi) · Demucs · speaker diarization & clustering (AHC, spectral) · TTS/STT · voice cloning · data augmentation

MLOps Prefect · MLflow · Docker / Docker Compose · torchrun & NCCL · multi-GPU distributed training · autoresume & checkpointing · CI

Cloud GCP (Vertex AI, GCS, Cloud Build, GCE) · AWS (S3, Lambda, IAM, access points) · Akamai · Wasabi · rclone

Backend Python · Flask · Node.js · PostgreSQL · MySQL · REST APIs

Frontend React · TypeScript · JavaScript · Next.js

AI providers ElevenLabs · Qwen/DashScope · OpenAI · Deepgram · Cohere · OpenVoice · DeepDub · Bhashini (Shrutam-2 / Sooktam-2)

EDUCATION & PUBLICATIONS

AISSMS College of Engineering, Pune — B.E. Computer Engineering, 8.85 CGPA

2020 – 2024

Undergraduate research: real-time Indian Sign Language recognition — custom Marathi dataset, MediaPipe keypoint extraction, BiLSTM sequence model.

- Indian Sign Language Recognition Using Holistic Body Pose Estimation and Deep Learning — Journal of Optoelectronics Technology, May 2024
- Indian Sign Language Recognition Using CNN Inception V3: A Review — IJRASET, Feb 2024